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## Questions

**Q1.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

In which of these esters is the percentage by mass of carbon 54.5%?

- ☒ **A** methyl methanoate, $HCOOCH_3$
- ☒ **B** methyl ethanoate, $CH_3COOCH_3$
- ☒ **C** ethyl ethanoate, $CH_3COOCH_2CH_3$
- ☒ **D** ethyl methanoate, $HCOOCH_2CH_3$

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**(Total for question = 1 mark)**

**Q2.**

This question is about ethanoic acid and some of its compounds.

(a) Three students, A, B and C, carried out an investigation to determine the amount of isoamyl acetate, an ester of ethanoic acid, in a banana food flavouring.

They hydrolysed the ester with excess sodium hydroxide solution.

$$CH_3COOCH_2CH_2CH(CH_3)_2 + NaOH \rightarrow CH_3COONa + (CH_3)_2CHCH_2CH_2OH$$

The amount of sodium hydroxide remaining was determined by titrating the reaction mixture with hydrochloric acid.

(i) Give the systematic name for isoamyl acetate.

**(1)**

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(ii) $6.06 \times 10^{-3}$ mol of sodium hydroxide reacted with the ester present in 1.07 g of the flavouring.
Calculate the percentage by mass of the ester in the food flavouring.

**(3)**

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(iii) The students used 25.0 $cm^3$ of sodium hydroxide solution with concentration 0.980 mol $dm^{-3}$ for the hydrolysis.
Calculate the pH of the reaction mixture when the hydrolysis is complete. Assume the total volume is

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25.0 $cm^3$.

[Ionic product of water, $K_w = 1.0 \times 10^{-14}\,mol^2\,dm^{-6}$]

**(3)**

(iv) Two indicators, methyl orange and phenolphthalein, were available for this titration.
Student A thought only methyl orange could be used.
Student B thought only phenolphthalein could be used.
Student C suggested either indicator could be used.
Explain which student is correct.

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**(2)**

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(v) Describe how the sodium ethanoate in the final reaction mixture could be converted into ethanoic acid.

**(1)**

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(b) Ethanoic acid can be used to form buffer solutions.

(i) Calculate the pH of the buffer solution formed when 50.0 $cm^3$ of ethanoic acid with concentration of 0.150 mol $dm^{-3}$ is mixed with 30.0 $cm^3$ of sodium hydroxide solution with concentration of 0.142 mol $dm^{-3}$.

[$K_a$ for ethanoic acid = $1.70 \times 10^{-5}\,mol\,dm^{-3}$]

**(5)**

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(ii) Small amounts of sodium hydroxide and hydrochloric acid are added to separate samples of the buffer solution in (b)(i).

Explain why these samples resist change in pH.

Illustrate your answer with at least two equations.

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**(5)**

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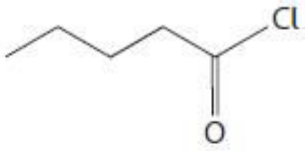
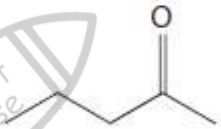
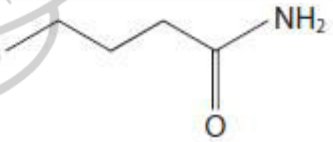
**(Total for question = 20 marks)**

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**Q3.**

This question is about four organic compounds each with five carbon atoms but different functional groups.

| Compound | Skeletal formula |
|---|---|
| 1 |  |
| 2 |  |
| 3 |  |
| 4 |  |

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(a) Which of these, when mixed with water, produces the solution with the **lowest** pH?

**(1)**

- ☒ **A** compound **1**
- ☒ **B** compound **2**
- ☒ **C** compound **3**
- ☒ **D** compound **4**

(b) Which of these reacts with ethylamine, $CH_3CH_2NH_2$, to form an N-substituted amide?

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**(1)**

- ☒ **A** compound **1**
- ☒ **B** compound **2**
- ☒ **C** compound **3**
- ☒ **D** compound **4**

(c) Which of these reacts with iodine and sodium hydroxide in solution to produce a pale yellow precipitate?

**(1)**

- ☒ **A** compound **1**
- ☒ **B** compound **2**
- ☒ **C** compound **3**
- ☒ **D** compound **4**

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**(Total for question = 3 marks)**

**Q4.**

A polymer is prepared by the reaction between hexanedioyl dichloride and hexane-1-6-diamine.

What type of polymerisation occurs?

- ☒ **A** addition
- ☒ **B** condensation
- ☒ **C** hydrolysis
- ☒ **D** substitution

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**(Total for question = 1 mark)**

**Q5.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

Which of the following could **not** be formed when methylamine is added to ethanoyl chloride?

- ☒ **A** $HCl$
- ☒ **B** $CH_3NH_3^+Cl^-$
- ☒ **C** $CH_3CONH_2$
- ☒ **D** $CH_3CONHCH_3$

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**(Total for question = 1 mark)**

**Q6.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

Butanoic acid may be prepared by the acid hydrolysis of

- ☒ **A** butyl ethanoate
- ☒ **B** 1-chlorobutane
- ☒ **C** ethyl butanoate
- ☒ **D** propanenitrile

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**(Total for question = 1 mark)**

**Q7.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

The formation of esters and the hydrolysis of esters are reactions that are slow under normal laboratory conditions.

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What speeds up these reactions?

| | Esterification | Hydrolysis |
|---|---|---|
| ☐ **A** | acids only | acids only |
| ☐ **B** | acids only | acids and bases |
| ☐ **C** | bases only | bases only |
| ☐ **D** | bases only | acids and bases |

**(Total for question = 1 mark)**

**Q8.**

**Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .**

Which reagent reacts at room temperature with methylamine, $CH_3NH_2$, to form the compound N-methylethanamide?

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☐ **A** $CH_3COCH_3$

☐ **B** $CH_3COOH$

☐ **C** $CH_3COOCH_3$

☐ **D** $CH_3COCl$

**(Total for question = 1 mark)**

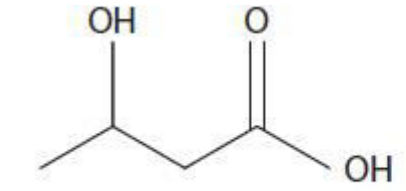
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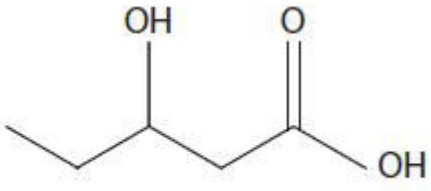
**Q9.**

**Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .**

The substance known as PHBV is a biodegradable polymer formed from 3-hydroxybutanoic acid and 3-hydroxypentanoic acid.




3-hydroxybutanoic acid



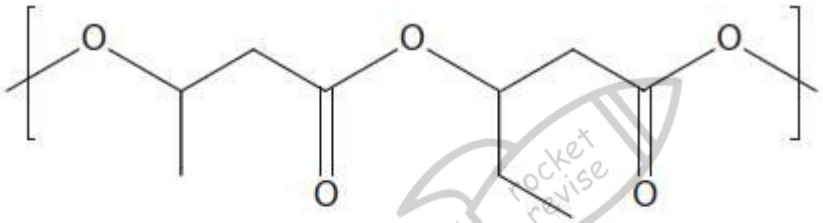

3-hydroxypentanoic acid

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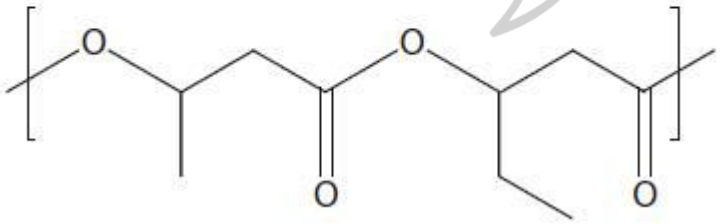
(a) Which of these is the repeat unit of the polymer?

(1)

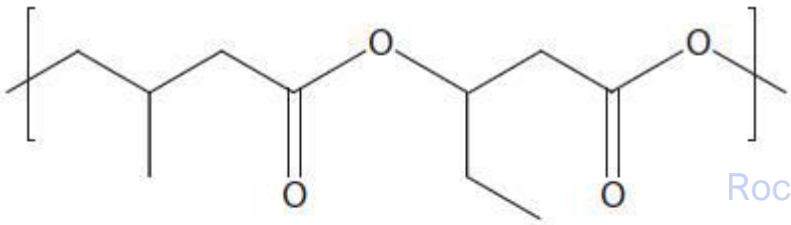
☐ **A**




☐ **B**

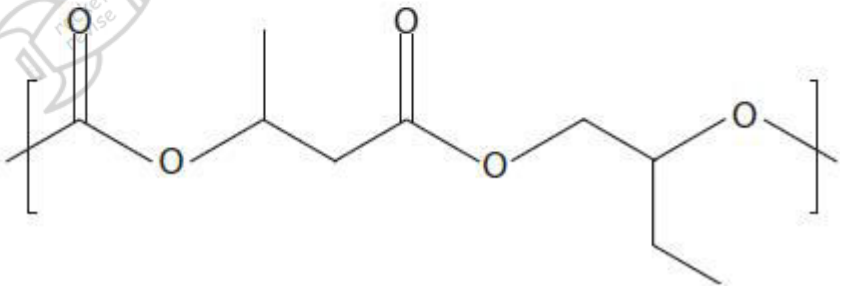



☐ **C**




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☐ **D**




(b) What reaction occurs when PHBV biodegrades to its monomers?

(1)

☐ **A** condensation

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☒ **B** hydrolysis
☒ **C** hydration
☒ **D** hydrogenation

**(Total for question = 2 marks)**

**Q10.**

**Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .**

Propyl ethanoate, $CH_3COOCH_2CH_2CH_3$ , is hydrolysed with aqueous sodium hydroxide.

Which products are formed?

☒ **A** $CH_3COOH$ and $CH_3CH_2CH_2OH$

☒ **B** $CH_3COOH$ and $CH_3CH_2CH_2ONa$

☒ **C** $CH_3COONa$ and $CH_3CH_2CH_2OH$

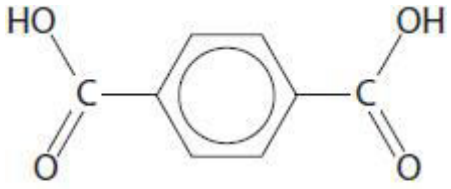
☒ **D** $CH_3COONa$ and $CH_3CH_2CH_2ONa$

**(Total for question = 1 mark)**

**Q11.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

Terylene is a polyester derived from ethane-1,2-diol and terephthalic acid.



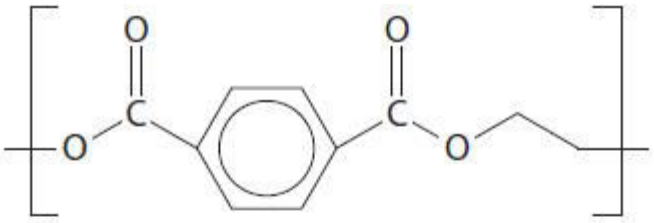

terephthalic acid

What is the structure of the repeat unit of terylene?

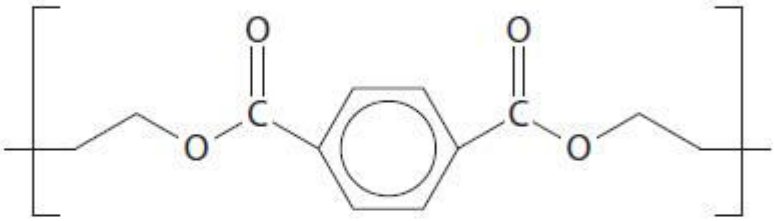
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☐ **A**

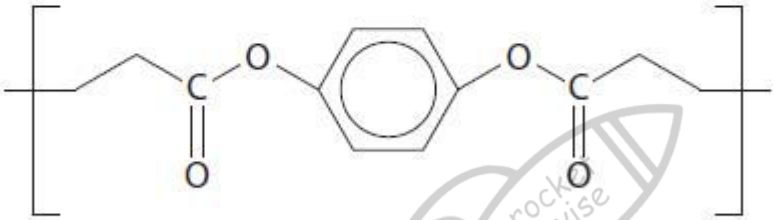


☐ **B**

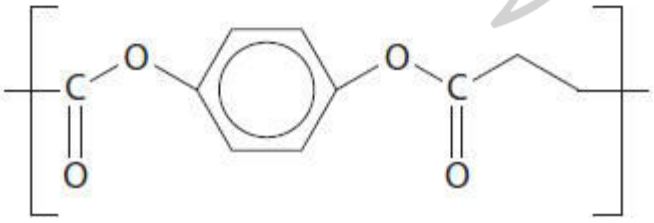


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☐ **C**



☐ **D**



**(Total for question = 1 mark)**

**Q12.**

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**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

Which compound is **both** completely miscible in water and has a higher boiling temperature than water?

☐ **A** $CH_3COOH$
☐ **B** $CH_3COCH_3$
☐ **C** $CH_3CH_2CH_2CHO$
☐ **D** $CH_3CH_2COOCH_2CH_3$

**(Total for question = 1 mark)**

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**Q13.**

Esters are used in flavourings and perfumes.
They can be made by reactions involving alcohols.

(a) A flask containing a mixture of 0.200 mol of ethanoic acid and 0.150 mol of ethanol was left at 25 °C, in the presence of a catalyst, until equilibrium had been established.

$$CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$$

The ethanoic acid present in the equilibrium mixture required 34.8 $cm^3$ of a 2.50 mol $dm^{-3}$ solution of sodium hydroxide for complete neutralisation.

(i) Calculate the value of the equilibrium constant, $K_c$, for this reaction at 25 °C.

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**(4)**

(ii) The enthalpy change for this reaction is small.
Explain, by reference to the type and number of bonds being broken and made, how this might have been predicted.

**(2)**

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(b) An ester which smells of raspberries can be formed by either of two different reactions.

Reaction **1**

$$HCOOH + (CH_3)_2CHCH_2OH \overset{\text{catalyst}}{\rightleftharpoons} HCOOCH_2CH(CH_3)_2 + H_2O$$

Reaction **2**

$$HCOCl + (CH_3)_2CHCH_2OH \rightarrow HCOOCH_2CH(CH_3)_2 + HCl$$

(i) **Name** the carboxylic acid, alcohol and catalyst used in Reaction **1**.

**(2)**

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HCOOH ..... catalyst .....

$(CH_3)_2CHCH_2OH$ .....

(ii) Explain **one** advantage and **one** disadvantage of using Reaction **2** rather than Reaction **1**.

**(4)**

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**(Total for question = 12 marks)**

**Q14.**

**For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

Which of these compounds is a product of the hydrolysis of $CH_3COOC_3H_7$ by refluxing with aqueous sodium hydroxide?

- ☐ **A** $CH_3OH$
- ☐ **B** $C_3H_7OH$
- ☐ **C** $C_3H_7COOH$
- ☐ **D** $C_3H_7COO^-Na^+$

**(Total for question = 1 mark)**

**Q15.**

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* Compare and contrast the reactions of ethanoyl chloride with water, with ethanol and with ammonia. Refer to the structural features of the molecules that determine the type of reaction. Identify the products of the reactions. You may include equations in your answers.

**(6)**

**(Total for question = 6 marks)**

**Q16.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

Which reaction does **not** produce a tertiary alcohol as one of the products?

- ☐ **A** $(CH_3)_2C=CH_2 + H_2O$
- ☐ **B** $(CH_3)_3CBr + H_2O$
- ☐ **C** $(CH_3)_3CCOC(CH_3)_3 + LiAlH_4$
- ☐ **D** $(CH_3)_3CCOOC(CH_3)_3 + NaOH$

**(Total for question = 1 mark)**

**Q17.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

Which reaction cannot form a carboxylic acid?

- ☐ **A** oxidation of primary alcohol
- ☐ **B** acidic hydrolysis of a nitrile
- ☐ **C** alkaline hydrolysis of an ester
- ☐ **D** acyl chloride reacting with water

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**(Total for question = 1 mark)**

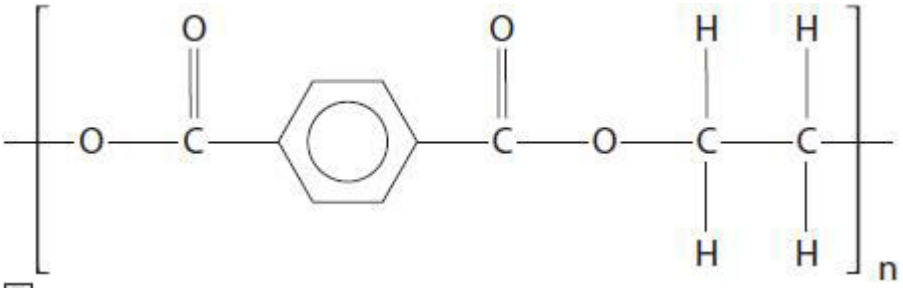
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**Q18.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

Which is produced after the polyester shown is hydrolysed with excess sodium hydroxide?



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- ☒ **A** benzene-1,4-dicarboxylic acid
- ☒ **B** ethane-1,2-diol
- ☒ **C** sodium ethanedioate
- ☒ **D** water

**(Total for question = 1 mark)**

**Q19.**

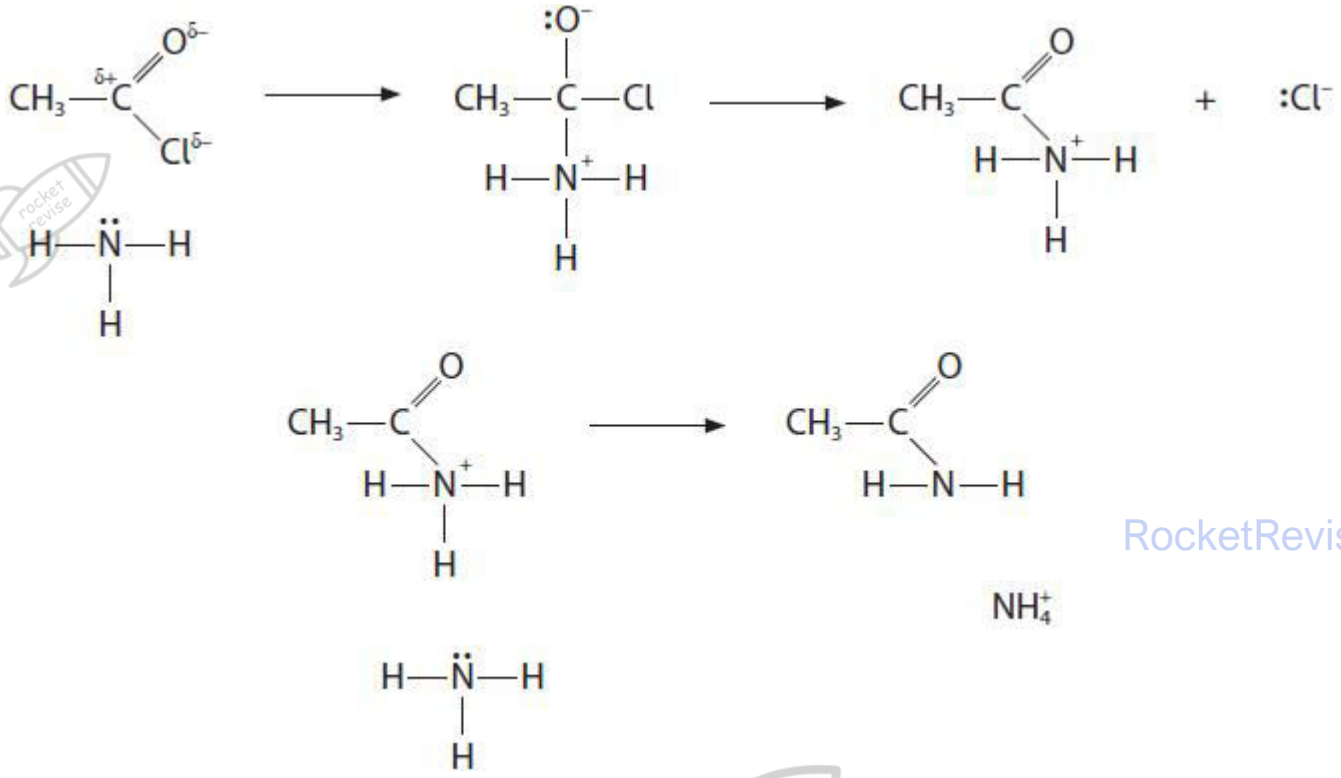
**Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

A partial mechanism of the reaction between ethanoyl chloride and concentrated aqueous ammonia is shown.

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(a) How many curly arrows are needed to complete the mechanism?

(1)

- ☒ **A** 4
- ☒ **B** 5
- ☒ **C** 6
- ☒ **D** 8

(b) What is the IUPAC name for the organic product?

(1)

- ☒ **A** amino ethanone
- ☒ **B** ethanamide
- ☒ **C** ethanoyl amine
- ☒ **D** methanamide

**(Total for question = 2 marks)**

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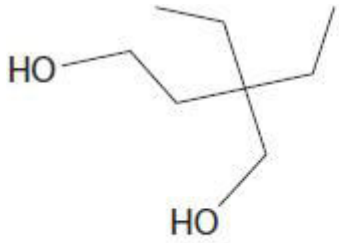
**Q20.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

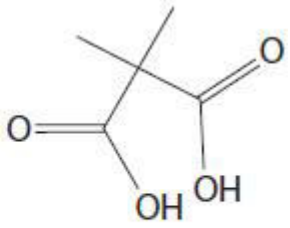
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The two monomers shown react to form a polymer.




monomer 1




monomer 2

Which is a repeat unit of the resulting polymer?

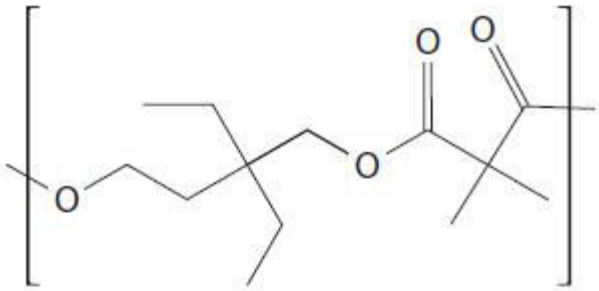
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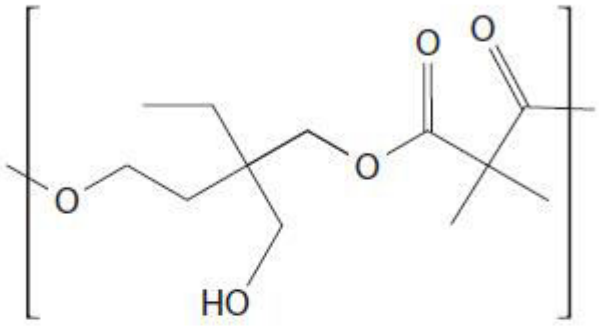
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☐ **A**

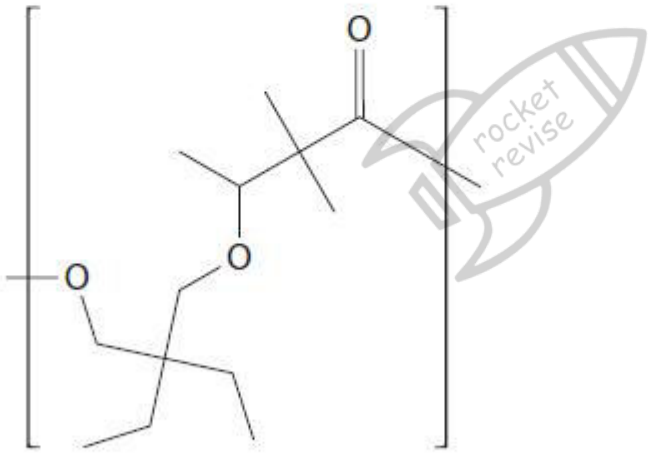


☐ **B**

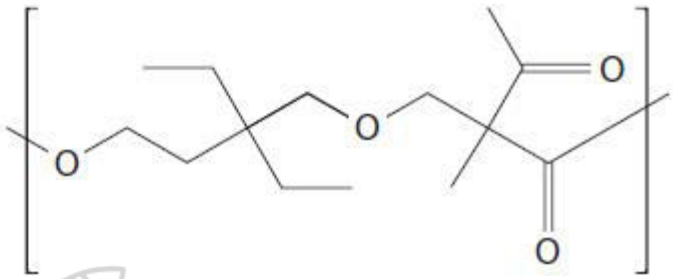



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☐ **C**



☐ **D**



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**(Total for question = 1 mark)**

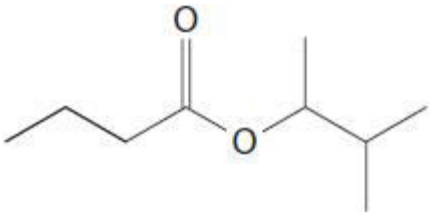
**Q21.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

The structure of an ester is shown.

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Which two compounds react to form this ester?

- ☒ **A** butanoyl chloride and 2-methylbutan-2-ol
- ☒ **B** butanoyl chloride and 3-methylbutan-2-ol
- ☒ **C** propanoyl chloride and 2-methylbutan-2-ol
- ☒ **D** propanoyl chloride and 3-methylbutan-2-ol

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**(Total for question = 1 mark)**

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**Q22.**

Polyesters can be made by the reaction of:

**(1)**

- ☒ **A** diprotic carboxylic acids with a primary alcohol
- ☒ **B** diprotic carboxylic acids with diols
- ☒ **C** monoprotic carboxylic acids with triols
- ☒ **D** monoprotic carboxylic acids with a secondary alcohol

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**(Total for question = 1 mark)**

**Q23.**

Carboxylic acids and acyl chlorides both react with alcohols to produce esters.
Acyl chlorides are often preferred to carboxylic acids for this reaction because the yield of the ester is usually higher.
Which of the following statements is the best explanation for the higher yield?

**(1)**

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- **A** acyl chlorides react to remove any water produced
- **B** the reaction is not reversible
- **C** the reaction is less exothermic
- **D** the reaction has a higher atom economy

**(Total for question = 1 mark)**

**Q24.**

Which reaction does **not** produce a carboxylic acid?

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**(1)**

- **A** hydrolysis of an acyl chloride with cold water
- **B** hydrolysis of an ester by refluxing with dilute hydrochloric acid
- **C** hydrolysis of a nitrile by refluxing with aqueous potassium hydroxide
- **D** oxidation of a primary alcohol by refluxing with excess acidified potassium dichromate

**(Total for question = 1 mark)**

**Q25.**

The organic product of the reaction of ethanoyl chloride with ammonia gas is:

**(1)**

- **A** ammonium ethanoate
- **B** ethanamide
- **C** ethanenitrile
- **D** methanenitrile

**(Total for question = 1 mark)**

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**Q26.**

(a) Ethyl propanoate, $CH_3CH_2COOCH_2CH_3$, smells of pineapple and is used as a flavouring. It may be hydrolysed using hydrochloric acid as a catalyst to produce propanoic acid and ethanol.

$$CH_3CH_2COOCH_2CH_3(l) + H_2O(l) \rightleftharpoons CH_3CH_2COOH(l) + CH_3CH_2OH(l)$$

A mixture was prepared using 0.100 mol of ethyl propanoate and 0.200 mol of water containing the catalyst.

The mixture was left to reach equilibrium at 25 °C.

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The equilibrium mixture contained 0.0440 mol of propanoic acid.

(i) Calculate the value for $K_c$ for this equilibrium at 25 °C.
Give your answer to an appropriate number of significant figures.

**(4)**

(ii) The standard enthalpy change, $\Delta_r H^{\ominus}$, for this reaction is close to, but not exactly zero. Explain this statement by considering the type and number of bonds being broken and made. No calculations are required.

**(2)**

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(iii) Deduce the effect of increasing the temperature on the total entropy change of this reaction, $\Delta S_{total}$, and on the value of the equilibrium constant, $K_c$. Assume that $\Delta S_{system}$ does not change with temperature.

**(3)**

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(b) Propanoic acid is a weak acid.

(i) State the difference between a weak acid and a strong acid such as hydrochloric acid.

**(1)**

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(ii) Calculate the pH of 0.500 mol $dm^{-3}$ hydrochloric acid at 25 °C.

**(1)**

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(iii) Calculate the pH of 0.500 mol $dm^{-3}$ propanoic acid at 25 °C.
$K_a$ (propanoic acid) = 1.30 × $10^{-5}$ mol $dm^{-3}$ at 25 °C.

**(3)**

(c) The number of moles of propanoic acid in a solution may be determined by titration with aqueous sodium hydroxide.

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(i) Calculate the pH at the point in the titration where half the acid has been neutralised. You must show your working.

**(2)**

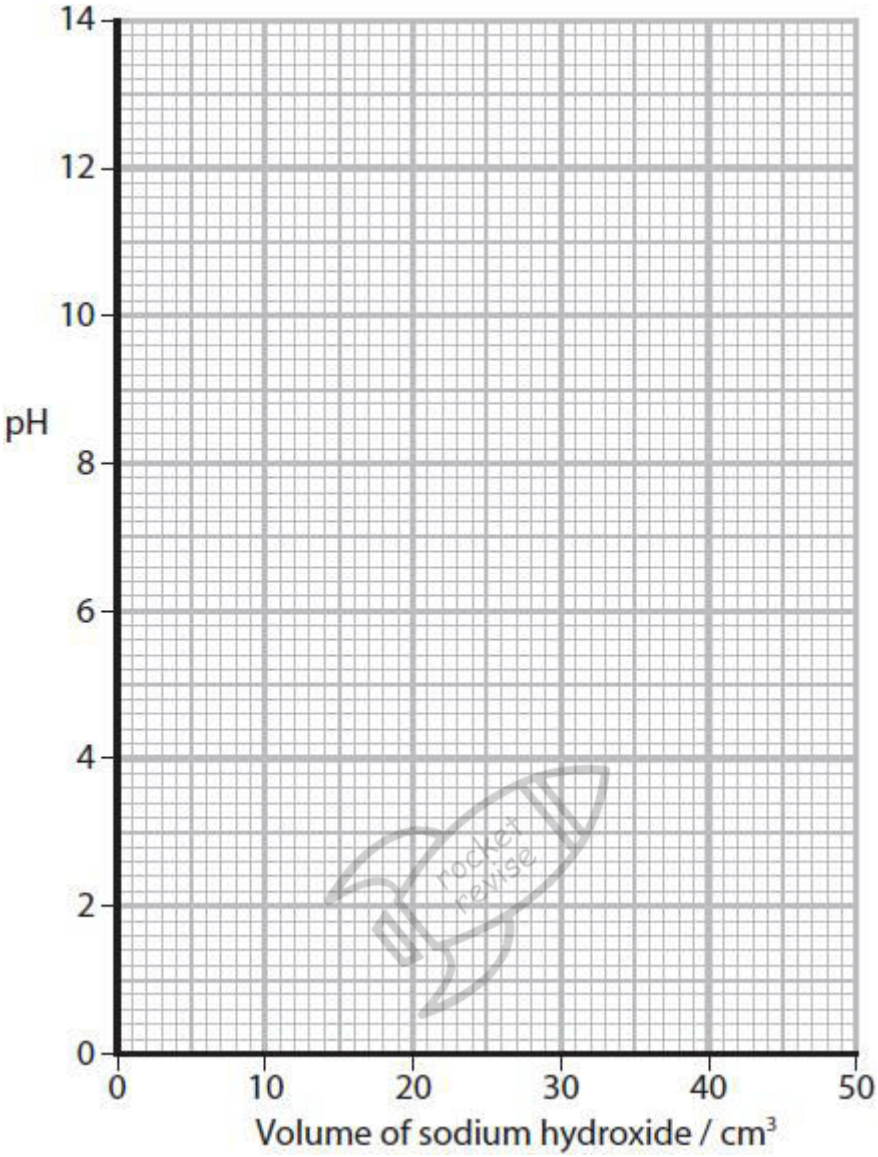
(ii) Sketch the titration curve showing the change in pH when 50 $cm^3$ of 0.500 mol $dm^{-3}$ sodium hydroxide is added to 25.0 $cm^3$ of propanoic acid of the same concentration, using your answers to (b)(iii) and (c)(i).

**(3)**

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(iii) Suggest a suitable indicator for this titration. Justify your choice by referring to your titration curve in (c)(ii).
Use the Data Booklet.

**(2)**

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**(Total for question = 21 marks)**

**Q27.**

**Answer the question with a cross in the box you think is correct (☒). If you change your mind**

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**about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).**

Which monomer could form a condensation polymer without needing a second type of monomer?

- ☐ **A** $HO(CH_2)_4OH$
- ☐ **B** $HO(CH_2)_4COCl$
- ☐ **C** $HO(CH_2)_4CH{=}CH_2$
- ☐ **D** $HO(CH_2)_4CH_2Cl$

**(Total for question = 1 mark)**

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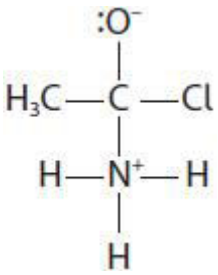
**Q28.**

**Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).**

Ethanoyl chloride reacts with ammonia, forming ethanamide.

$$CH_3COCl + NH_3 \rightarrow CH_3CONH_2 + HCl$$

The first step of the mechanism forms the intermediate shown.



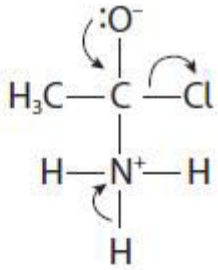

Which diagram shows the movements of electron pairs when ethanamide forms from this intermediate?

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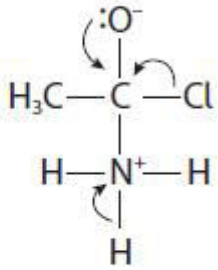
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☒ **A**

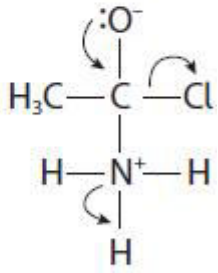


☒ **B**

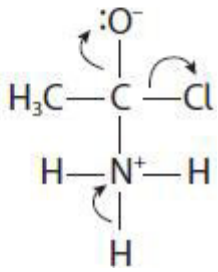


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☒ **C**



☒ **D**



**(Total for question = 1 mark)**

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**Q29.**

**Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).**

Which statement(s) is/are correct?

$CH_3CH_2COCl$ reacts with

1 $CH_3NH_2$ to form $CH_3CH_2CONHCH_3$
2 $(CH_3)_2NH$ to form $CH_3CH_2CON(CH_3)_2$
3 $(CH_3)_3N$ to form $CH_3CH_2CON(CH_3)_3$

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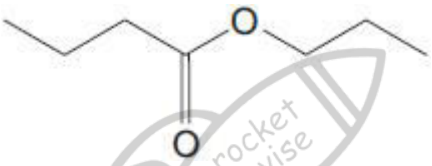
- ☒ **A** 1, 2 and 3
- ☒ **B** 1 and 2 only
- ☒ **C** 2 and 3 only
- ☒ **D** 1 only

**(Total for question = 1 mark)**

**Q30.**

**Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).**

An ester has the structure shown.




Which pair of reactants would form this ester, under suitable conditions?

- ☒ **A** butanoic acid and propan-2-ol
- ☒ **B** butanoyl chloride and propan-1-ol
- ☒ **C** propanoic acid and butan-1-ol
- ☒ **D** propanoyl chloride and butan-2-ol

**(Total for question = 1 mark)**

**Q31.**

**Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).**

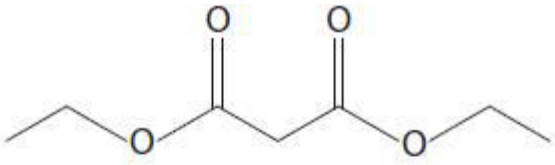
An ester is hydrolysed in excess dilute hydrochloric acid to form ethanol and propanedioic acid.

What is the skeletal formula of this ester?

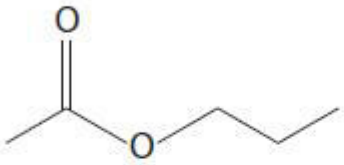
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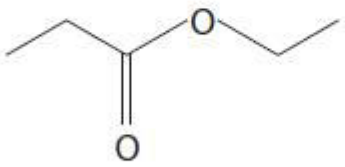
☐ **A**



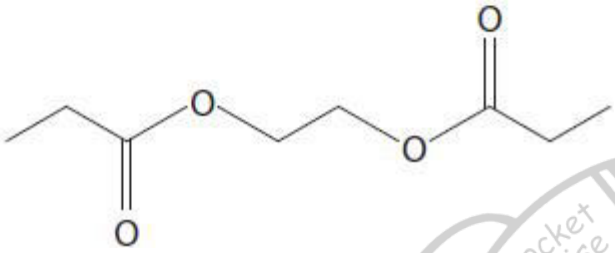
☐ **B**



☐ **C**



☐ **D**



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**(Total for question = 1 mark)**

**Q32.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

From which two reactants could $CH_3(CH_2)_2COO(CH_2)_4CH_3$ be made?

☐ **A** butanoic acid and pentan-1-ol
☐ **B** butanoyl chloride and butan-1-ol
☐ **C** butanal and pentan-1-ol
☐ **D** pentanoic acid and butan-1-ol

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**(Total for question = 1 mark)**

**Q33.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

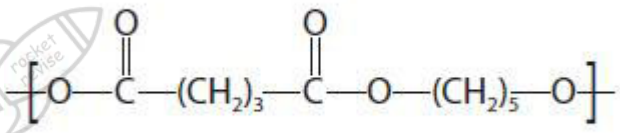
A polyester is made from pentane-1,5-diol and pentanedioic acid. Which is the repeat unit of this

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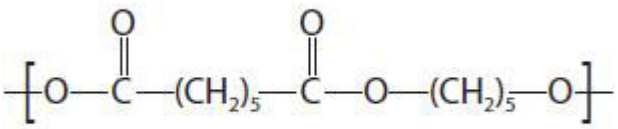
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polyester?

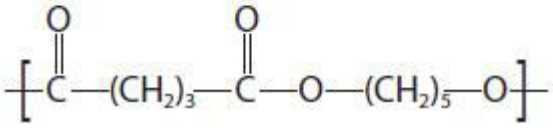
☒ **A**



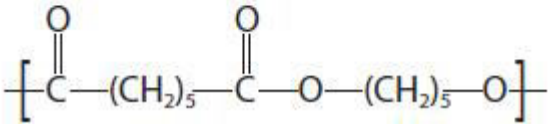
☒ **B**



☒ **C**



☒ **D**



**(Total for question = 1 mark)**

**Q34.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

Which of these statements about the reaction of ethanoyl chloride with ammonia is **not** correct?

- ☒ **A** an amide is formed
- ☒ **B** the reaction is exothermic
- ☒ **C** misty fumes and white smoke are given off
- ☒ **D** a gas is produced that turns damp red litmus paper blue

**(Total for question = 1 mark)**

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